

ASSIGNMENT 7 REPORT

Secure Network Application Development Using TCP

Name: Aman

Roll no.: 241001160007

1. Objective

The objective of this assignment is to enhance the GUI-based multi-client TCP chat application from Assignment 6 by implementing practical security mechanisms including user authentication, SHA-256 password hashing, duplicate login prevention, failed login protection, session management, secure logging, input validation and Wireshark verification.

2. Software and Tools

- Python 3
- Tkinter
- Socket Programming
- Mininet
- Wireshark
- CSV
- hashlib (SHA-256)

3. System Architecture

The application follows a client-server architecture. Clients authenticate with the TCP server on port 5000. The server validates credentials, manages active sessions, prevents duplicate logins, logs security events and broadcasts messages to connected users.

4. Security Features Implemented

- Username and password authentication.
- Passwords stored as SHA-256 hashes in users.csv.
- Automatic registration of new users with hashed passwords.
- Duplicate login prevention.
- Input validation for usernames, passwords, commands and message length.
- Temporary account lock after five failed login attempts.
- Session timeout after 180 seconds of inactivity.
- Secure logging in security_log.txt without storing plaintext passwords.

5. Implementation

- The client validates user input before connecting.
- The server hashes passwords using `hashlib.sha256()` before storage and verification.
- The server maintains active sessions and rejects duplicate logins.
- Unsupported commands and oversized messages are rejected.

- Security events such as login success, login failure, registration, logout and session timeout are recorded.

6. Testing

Test Case	Expected Result	Status
Valid Login	Authentication successful	Pass
Incorrect Password	Access denied	Pass
Duplicate Login	Second login blocked	Pass
Failed Login Attempts	Temporary lockout	Pass
Oversized Message	Rejected	Pass
Session Timeout	Automatic logout	Pass

7. Wireshark Verification

Network traffic was captured using the filter tcp.port == 5000. Successful authentication, failed authentication, authenticated communication and logout/session termination were verified from the captured TCP packets.

8. Screenshots

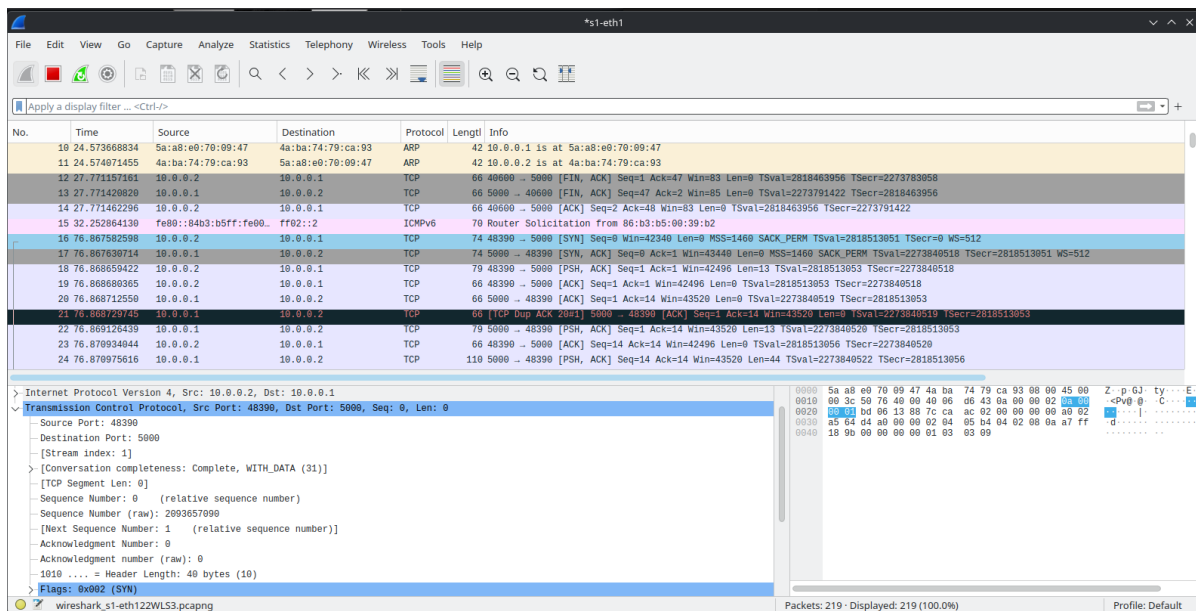


Figure 1: Login Window

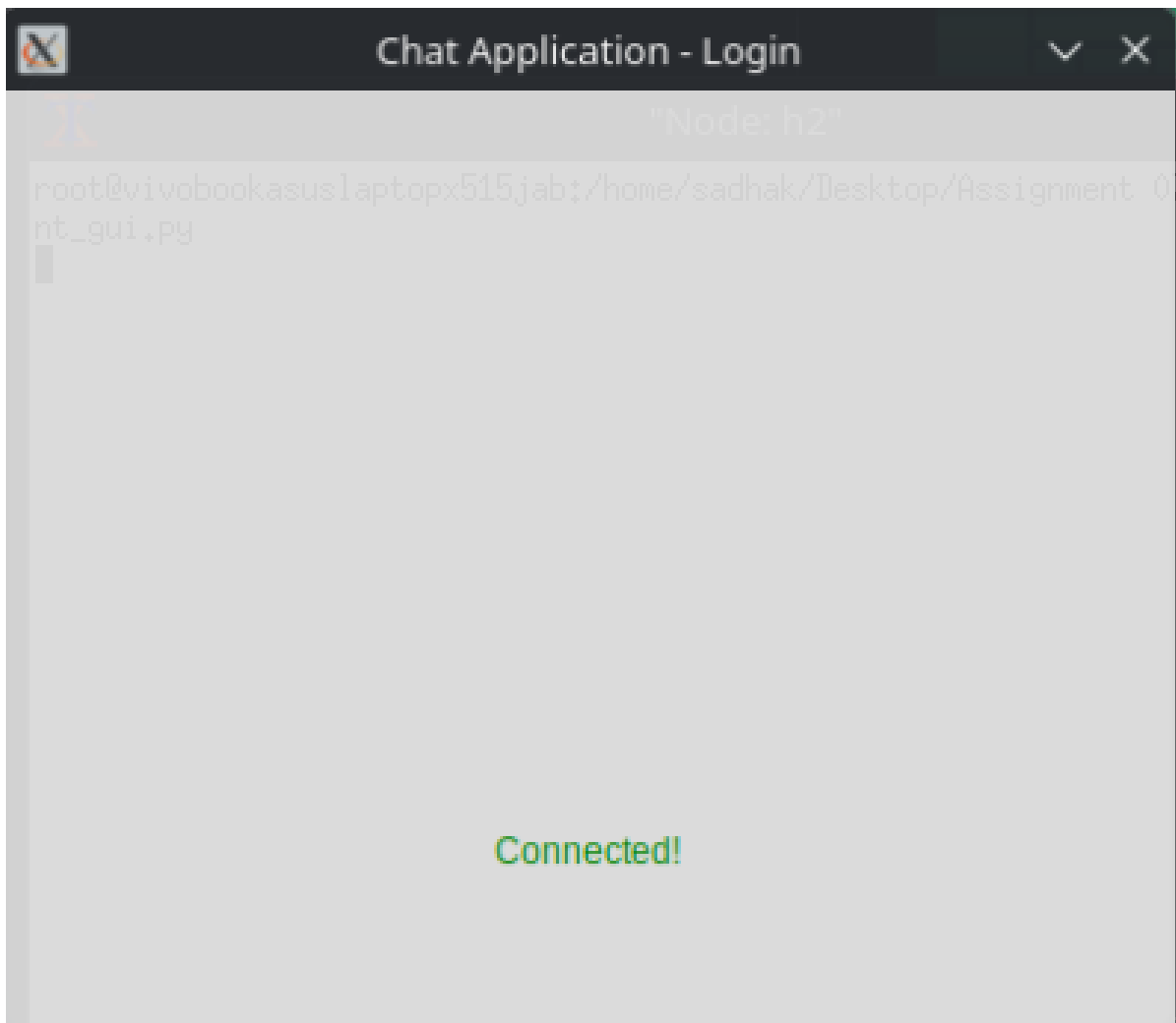


Figure 2: Successful Authentication

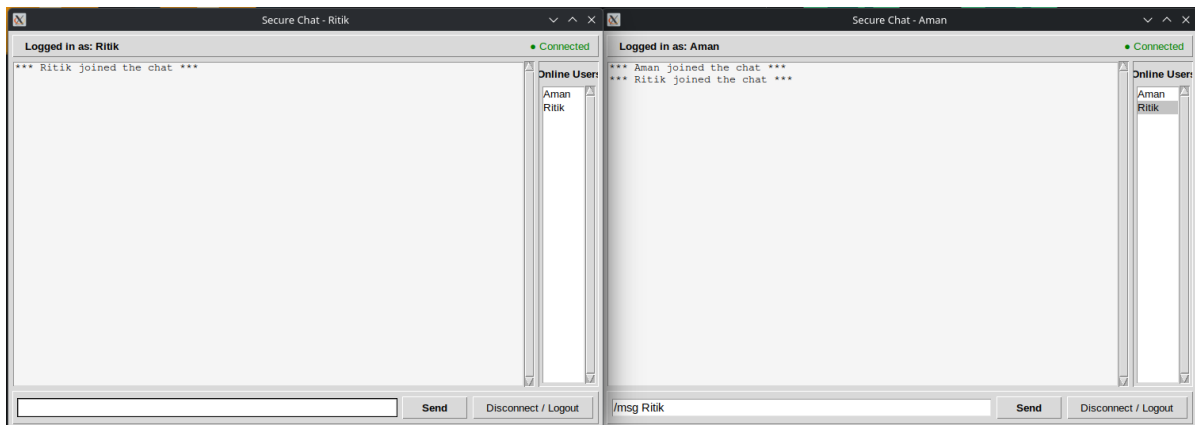


Figure 3: Chat Interface

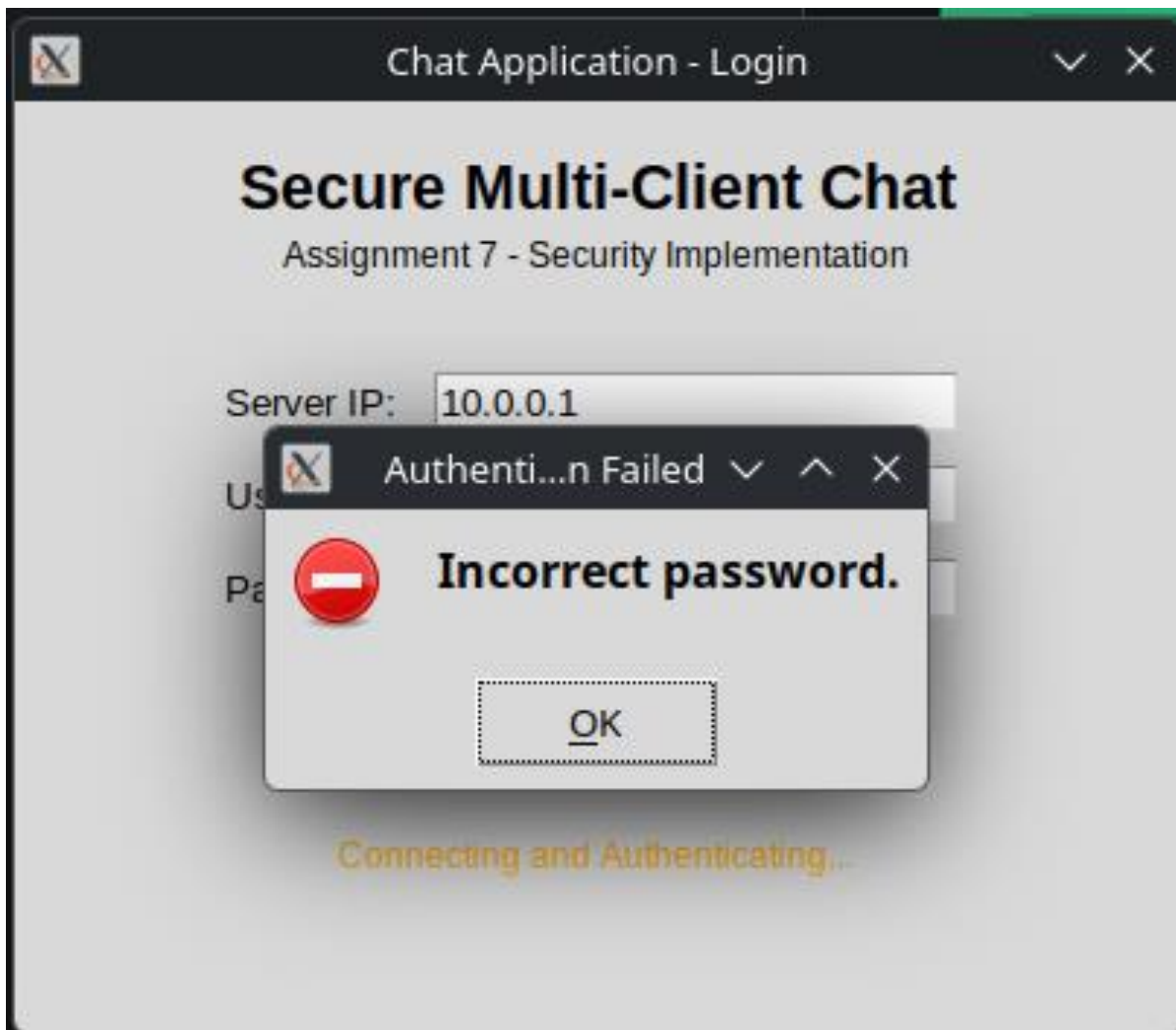


Figure 4: Incorrect Password

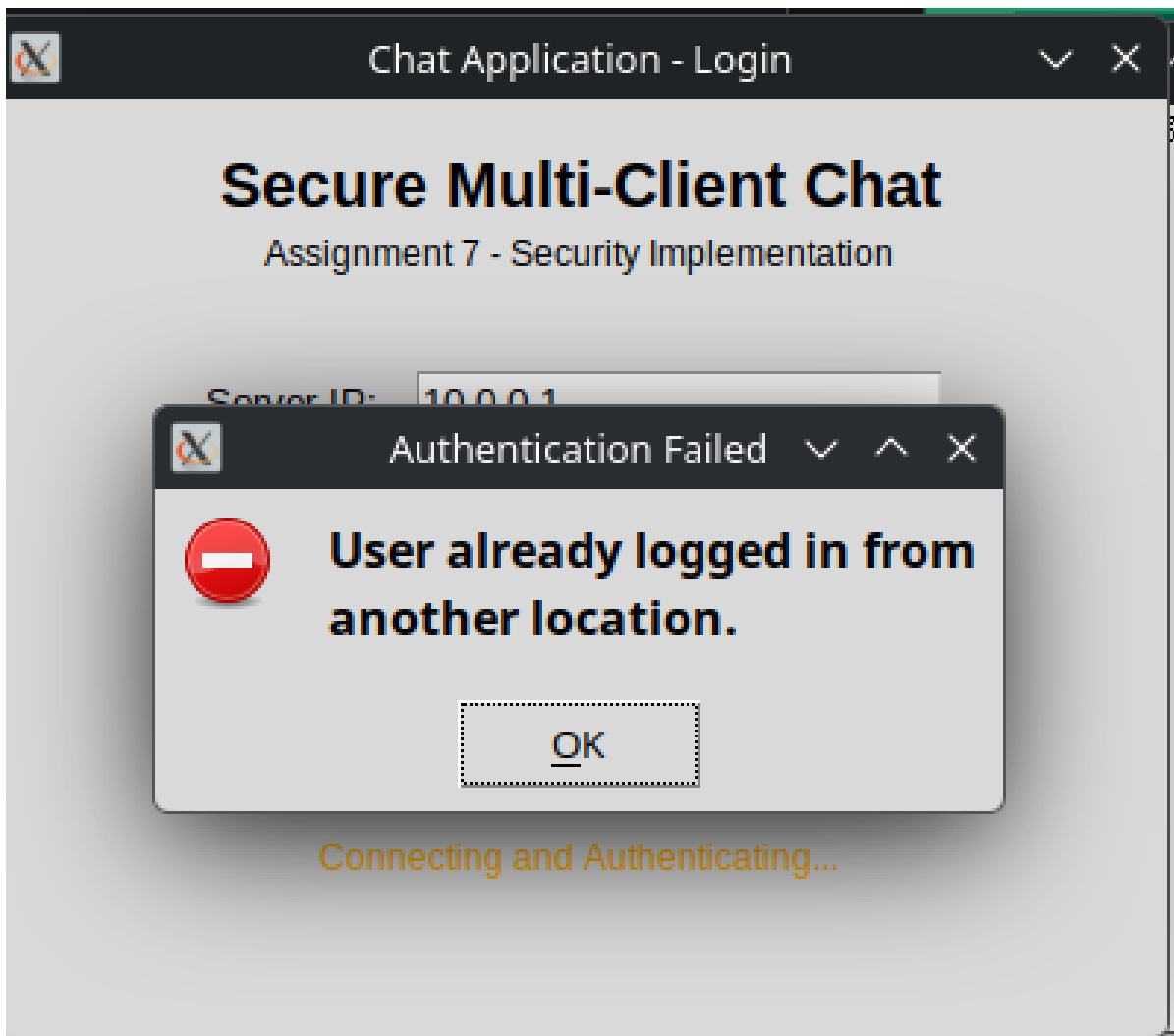


Figure 5: Duplicate Login Prevention

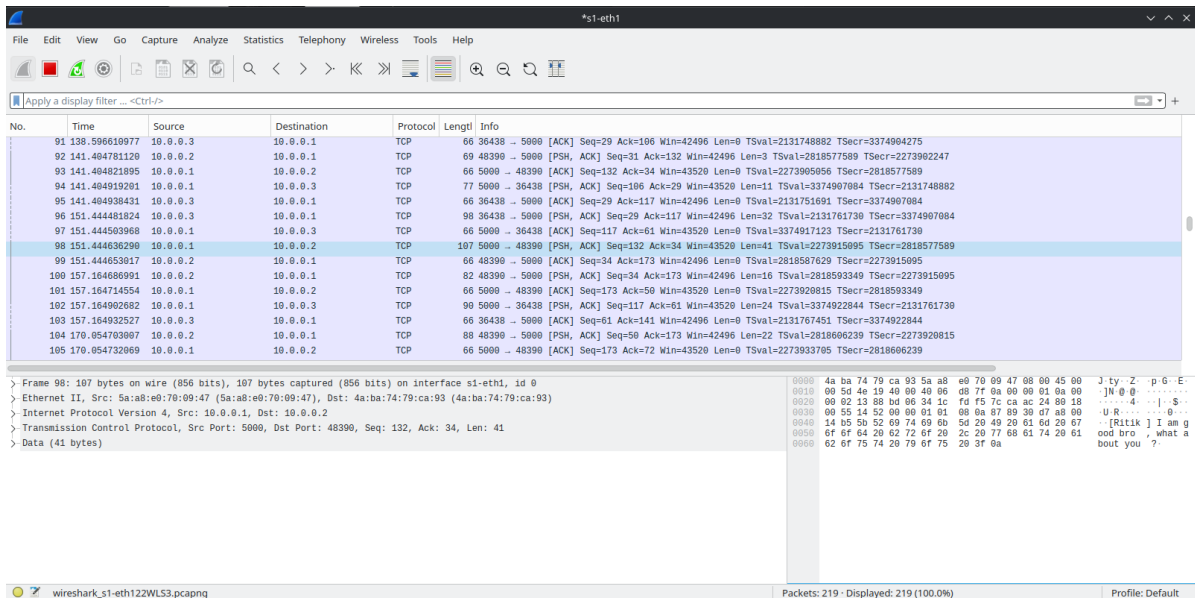


Figure 6: Authenticated Communication

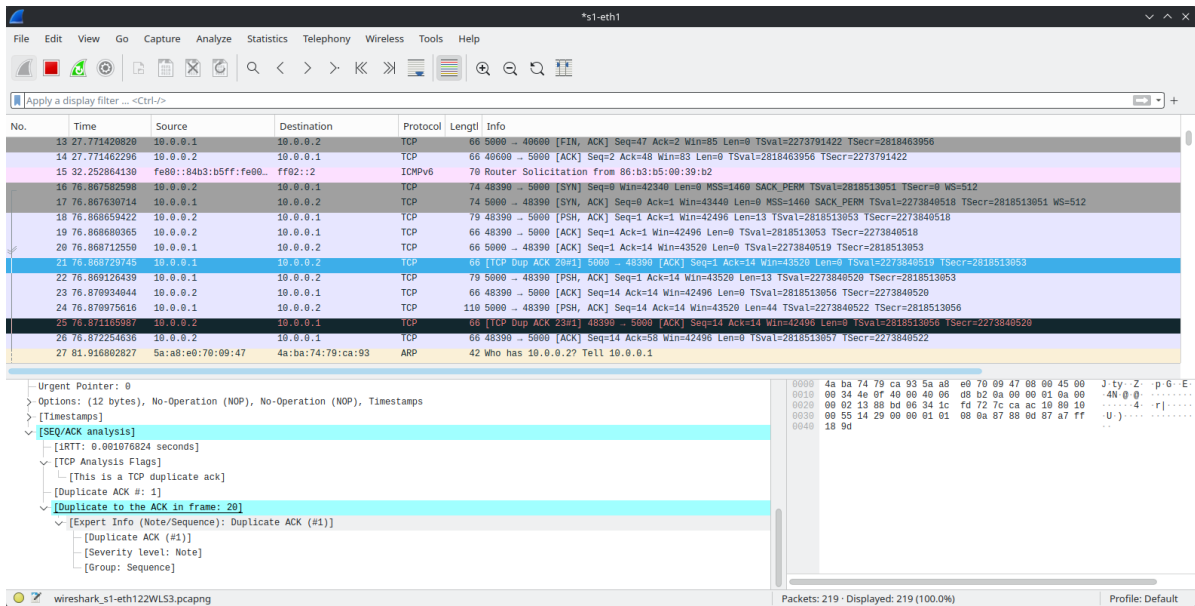


Figure 7: Failed Login in Wireshark

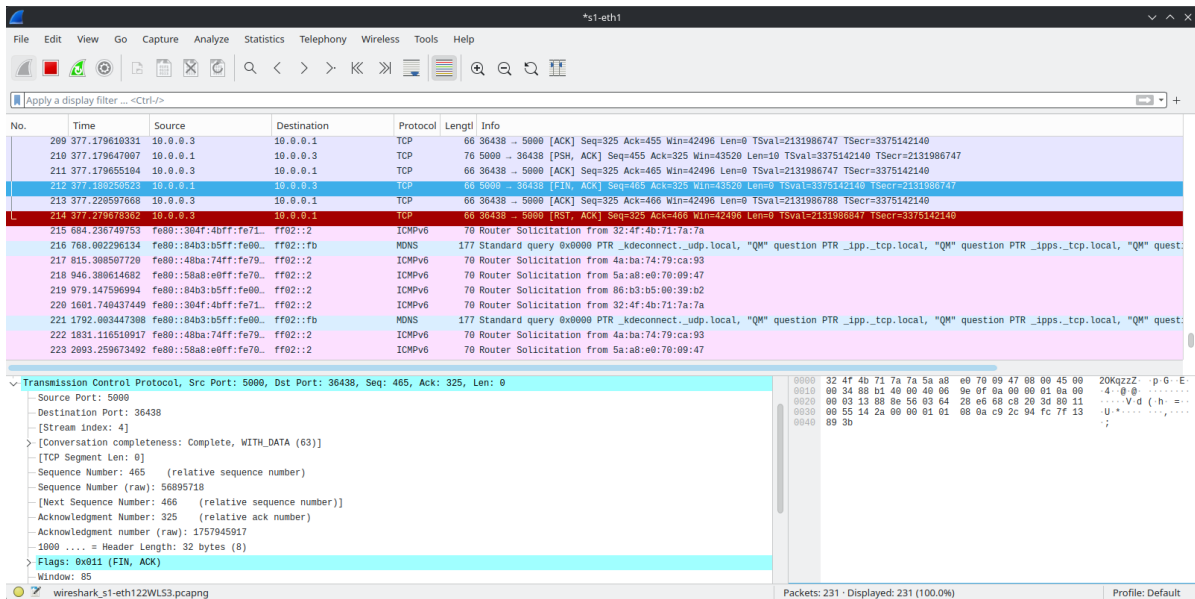


Figure 8: Logout / Session Disconnect

9. GitHub Repository

A new public GitHub repository was created for Assignment 7 to maintain a clean and organized project structure. The implementation reuses the networking and GUI code developed in Assignment 6 and extends it by adding the required security features, including user authentication, SHA-256 password hashing, duplicate login prevention, failed login protection, input validation, session management, secure logging, and Wireshark verification.

10. Conclusion

The secure TCP chat application successfully implements all required security mechanisms. Authentication, password hashing, duplicate login prevention, session management, secure logging and input validation significantly improve application security while maintaining usability.

Handwritten Reflection Questions

Ques-1. Explain the difference between authentication and authorization.

Answer: Authentication is the process of verifying a user's identity using a username and password. Authorization determines what an authenticated user is allowed to access or do after logging in.

Ques-2. Why should passwords never be stored in plaintext?

Answer: Passwords should never be stored in plaintext because anyone who gains access to the database can read them. Storing hash passwords protect users' accounts even if the database is compromised.

Ques-3. What is the purpose of the password hashing?

Answer: Password hashing converts a password into a fixed-length encrypted hash that cannot easily be reversed. During login, the entered password is hashed and compared with the stored hash to verify the user securely.

Ques-4. How does duplicate login prevention improve security?

Answer: Duplicate login prevention stops the same account from being used on multiple devices at the same time. This reduces the risk of unauthorized access, account sharing, and session hijacking.

Ques-5. Suggest two additional security improvement for your application.

Answer:

1. Implement Two-Factor Authentication (2FA) to require a second verification step during login.
2. Encrypt all communication between the client and server using SSL/TLS to protect data from interception.